

Activity Dossier

Lab 3 — Dashboard Project

Build a rail KPI dashboard in Looker Studio ($\approx$ 1 hour)

Student package — everything you need to do this activity

Field	Detail
Module	Business Intelligence & KPIs
Builds on	Your KPI / data-storytelling lecture, and the pandas EDA from Labs 1–2
Duration	$\approx$ 1 hour
Tool	Google Looker Studio — free, in the browser (needs a free Google account)
Data	Real UK rail station usage 2024–25 (Office of Rail and Road)
Hand in	Your dashboard (PDF or share link) + a short written insight note

1. The situation

You are an analyst at a rail operator. Leadership doesn't want a spreadsheet — they want a one-screen dashboard that answers, at a glance: how busy is the network, which stations and regions carry the most passengers, and which operators run the busiest stations. Your job is to turn a raw data table into a clear KPI dashboard, and then explain what it says.

A KPI (Key Performance Indicator) is a single number that tells you how something is doing — here, things like total passenger entries + exits, the number of stations, and the busiest station. A good dashboard shows a few well-chosen KPIs plus a couple of charts, all on one screen.

2. What you will be able to do

- Load a real data table into a business-intelligence tool (Looker Studio).
- Create KPI scorecards for the headline numbers.
- Build bar charts and a ranked table to compare regions, operators and stations.
- Arrange it into a clean, one-screen dashboard and share it.
- Write a short insight note that turns the dashboard into a recommendation.

3. What's in your package

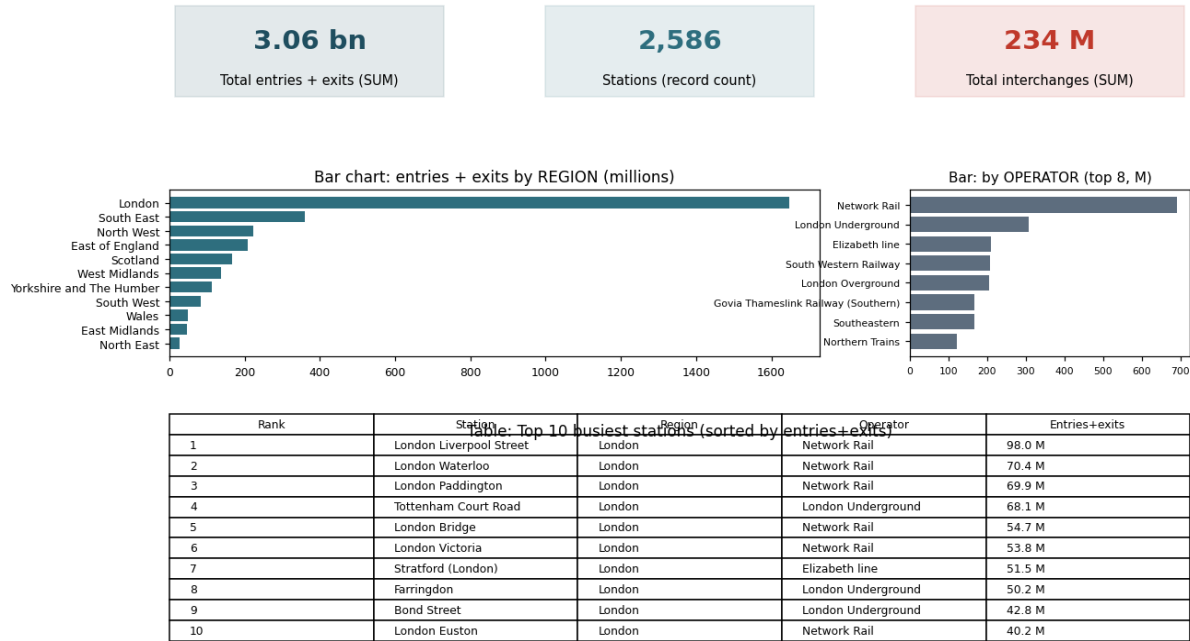
File	What it is
uk_station_usage_2024_25.csv	The dataset you upload to Looker Studio.

File	What it is
reference_dashboard.png	A picture of the finished dashboard you are aiming for.
Student_Insight_Note_Lab3.docx	Where you WRITE your findings — the second deliverable.
Activity_Dossier_Lab3.pdf	This document — your step-by-step guide.
START_HERE.txt	One-page quick start.

4. What you're aiming for

Here is the finished dashboard you will build — three KPI scorecards, a bar chart by region, a bar chart by operator, and a top-10 stations table, all on one page:

UK Rail Station Usage 2024-25 — KPI Dashboard (Looker Studio target)



5. Before you start

1. Open lookerstudio.google.com in your browser and sign in with a free Google account.
2. Keep the file `uk_station_usage_2024_25.csv` handy — you will upload it in Step 2.

Data note: this is real, open UK government data (Office of Rail and Road). It is public and free to use — there is nothing sensitive here.

6. Build the dashboard — step by step

Follow these in order. Each is a few clicks. Don't worry about making it pretty until the end.

Step 1 — Create a blank report

3. On the Looker Studio home page, click **Create** ► **Report** (or “Blank report”).
4. If asked, accept the terms and country settings.

Step 2 — Connect your data

5. In the “Add data to report” panel, find and click the “File Upload” connector.
6. Click **Upload files**, choose `uk_station_usage_2024_25.csv`, wait for it to finish, then click **Add**.
7. If a pop-up asks “Add data to this report?”, click **Add**. Your table is now connected.

Step 3 — Check the fields (dimensions vs metrics)

Looker Studio splits your columns into **dimensions** (categories — the green fields like `region`, `operator`, `station_name`) and **metrics** (numbers — the blue fields like `entries_exits_total`). You'll use both.

Step 4 — Add three KPI scorecards

A scorecard shows one big number. Add three:

8. Click **Add a chart** ► **Scorecard**, draw it on the page. Set its **Metric** to `entries_exits_total` (it will show ≈ 3.1 billion). Rename it “Total entries + exits”.
9. Add another Scorecard; set the Metric to **Record Count** (shows ≈ 2,586 — the number of stations). Label it “Stations”.
10. Add a third Scorecard; Metric = `interchanges`. Label it “Total interchanges”.

Tip: if a metric shows an average instead of a total, click the metric and set its aggregation to **SUM**.

Step 5 — Bar chart: usage by region

11. Add a chart ► **Bar chart**. Draw it below the scorecards.
12. Set **Dimension** = `region` and **Metric** = `entries_exits_total`.
13. In the chart's **SORT** setting, sort by `entries_exits_total`, descending. (London will be far the largest.)

Step 6 — Bar chart: top operators

14. Add another Bar chart. Dimension = `operator`, Metric = `entries_exits_total`, sorted descending.
15. Set “Bars” (number of bars) to 10 so you see the top 10 operators.

Step 7 — Table: the 10 busiest stations

16. Add a chart ► Table. Draw it across the bottom.
17. Dimensions = station_name, region, operator. Metric = entries_exits_total. Sort descending, and set “Rows per page” to 10.

Step 8 — Title and tidy up

18. Add a text box at the top: “UK Rail Station Usage 2024–25 — KPI Dashboard”.
19. Line the charts up on one page. Optionally pick a Theme (Theme and layout ► Theme).

Step 9 — Make it YOURS (your individual part)

So far everyone's dashboard is the same. Now make yours different: choose ONE region or operator you find interesting and add a view focused on it.

20. Pick your region or operator (your choice — e.g. Scotland, or South Western Railway). Then add **one new Scorecard** and **one new Bar chart** for it.
21. On each new chart, open **Setup ► Add a filter ► Create a filter**, set Include region (or operator) equal to your chosen one, and Save. (Or add a Control ► Drop-down list on region and select yours to filter the whole page.)
22. Write down which region/operator you chose — you will discuss it in your insight note.

Why this matters: your chosen slice + your written note are the parts of the submission that are uniquely yours — they can't be copied from the live demo. Everyone builds the same base; you make it your own here.

Step 10 — Share / export (submit this)

23. Click **File ► Download report** and save as PDF — OR click **Share** and copy the view link.

7. The second deliverable — your insight note

A dashboard is not the whole answer. Open Student_Insight_Note_Lab3.docx and write, in your own words, the 3–4 things your dashboard reveals and one recommendation for the operator. **Do not just paste screenshots or numbers — explain what they MEAN.** A note that is only pictures/numbers earns no marks.

8. What to hand in & how it's graded (100 marks)

Hand in TWO things: your dashboard (PDF or share link) — including your personalised view from Step 9 — and your written insight note.

Component	Marks	Full marks means
Base dashboard built correctly	30	3 scorecards, 2 bar charts, top-10 table, one page, titled
Your personalised view (Step 9)	15	A region/operator YOU chose, shown with a filter + a chart/scorecard
Insight note: what the data means	35	3–4 findings incl. your chosen region vs the national picture, in your own words
Insight note: one recommendation	20	A sensible, data-backed action for the operator

9. Your data — where it comes from

UK rail station usage, April 2024–March 2025 (entries, exits and interchanges at every station in Great Britain), from the Office of Rail and Road, Table 1410. One row per station, with its region, operator and ticket-type breakdown.

Licence: Open Government Licence v3.0 — free to use, including commercially, with attribution. It is provided in your package; you do not need to download anything.

10. Getting unstuck

- Can't find File Upload? In “Add data”, scroll the connectors list — it's under “Connect to data”, Google connectors.
- A number looks tiny or is an average? Click the metric and set aggregation to SUM.
- London swamps every chart? That's real — note it in your insight note; optionally add a filter to exclude London and compare.
- Chart empty? Check you put a metric (blue) in the Metric box and a dimension (green) in the Dimension box.